

TMR 4190 FINITE ELEMENT METHODS IN STRUCTURAL ANALYSIS

AUTUMN 2026

EXERCISE 5

Problem 1



Figure 1

A preliminary design study of a crane for offshore applications needs to be performed. The structure is schematized as a planar frame consisting of several beam elements. Figure 2 shows geometry (dimensions in meters), loading and boundary conditions for the crane model.

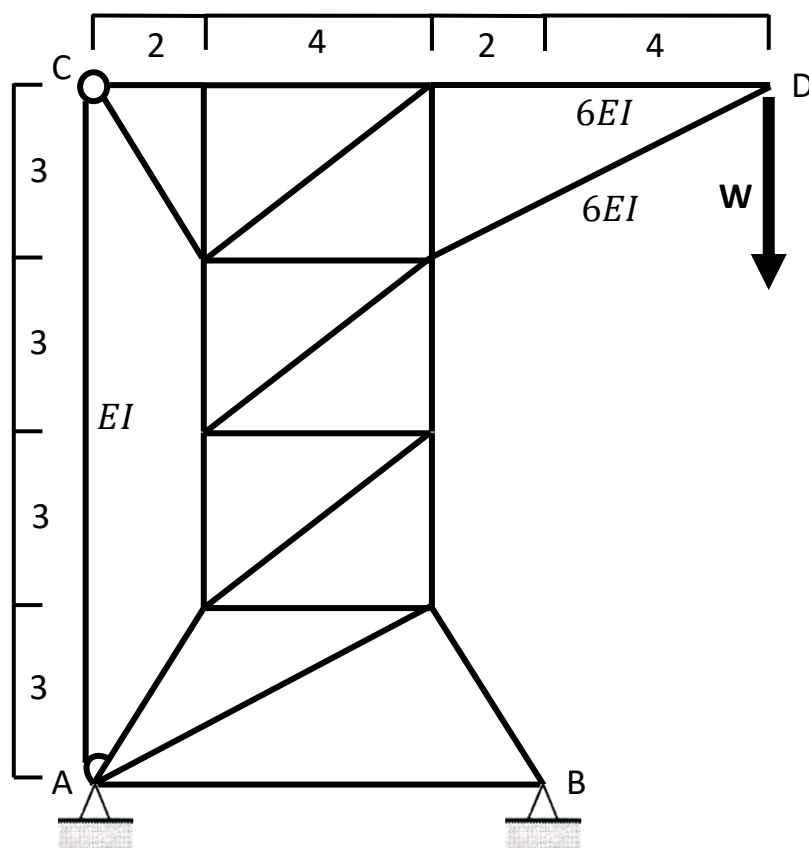


Figure 2

The beam elements are mostly rigidly connected to each other, but internal hinges are present at the points A and C. Note that, in C, the rotations of all the three connected members are independent, while in B only the vertical element has independent rotation with respect to the other three beams (that are rigidly connected to each other). The structure is connected to the basement by simple supports that prevent the displacements (not rotations) of the points A and B. All the beams, if not specified differently in Figure 2, have bending stiffness $3EI$ and axial stiffness EA . The crane is designed for lifting a weight W at point D.

The problem parameters are given as follows:

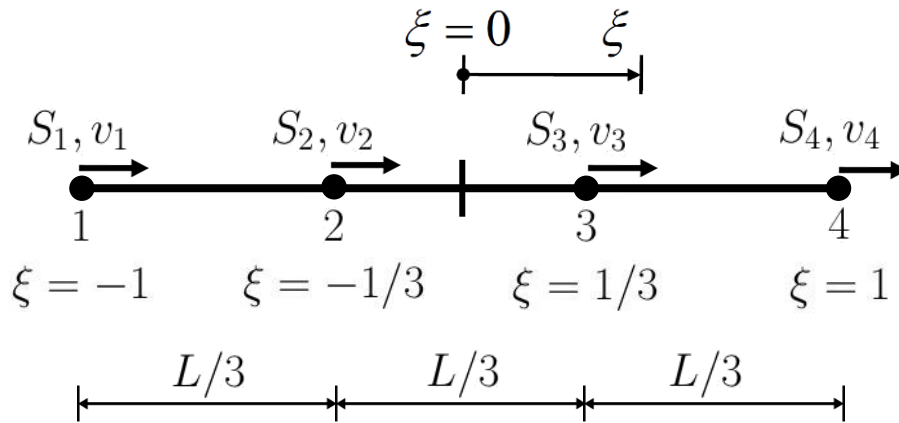
$$EI = 10^4 \text{ Nm}^2, EA = 3 \times 10^9 \text{ N}, W = 200\,000 \text{ N}.$$

- a) Using the Python/Matlab implementation of the Matrix Method and imposing the boundary conditions via the fixed dof vector (as in example problem 5d), compute the displacement vector of the structure $\mathbf{r}$. How much is δ , the vertical displacement of the point to which the weight is attached?
- b) Compute the reactions forces in the supports of the structure, and check if the equilibrium between the reactions in A, B and the external force is satisfied.
- c) Determine the internal forces (N, V, M) in the element A-C.

Include in your solutions the numerical results and the discussions required, the vector $\mathbf{r}$ and the plot of the deformed geometry. Upload also the main script (as .m file) that you run for the simulation.

Problem 2

Compute the Lagrange polynomials for the cubic four-node bar element having equally spaced nodes, see figure below.



Due date for Exercise 5 -.